Exploration & Mitigation of Gender Bias in Word Embeddings from Transformer-based Language Models

Ariyan Hossain, 20101099
Rakinul Haque, 20101290
Khondokar Mohammad Ahanaf Hannan, 20101079
Nowreen Tarannum Rafa, 20101329
Humayra Musarrat, 20101089

Supervisor: Dr. Farig Yousuf Sadeque



Introduction

- An important societal problem that maintains negative perceptions and disparities between men and women is gender bias.
- Machine learning (ML) has the potential to reveal biases in information caused by human mistake when it is used without the right amount of caution.
- The core problem comes from word embedding, which is a popular way to use vectors to represent text data in machine learning and natural language processing tasks.

Problem Statement

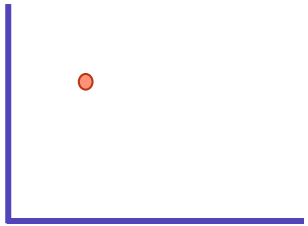


How can we detect gender bias in contextualized word embeddings?

How can we effectively mitigate gender bias from these models?

Word Embeddings

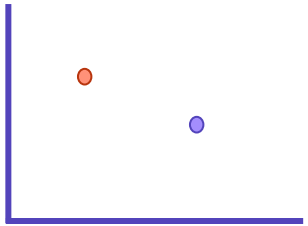
Static



I watch TV
I wear watch

Glove, Word2Vec, etc

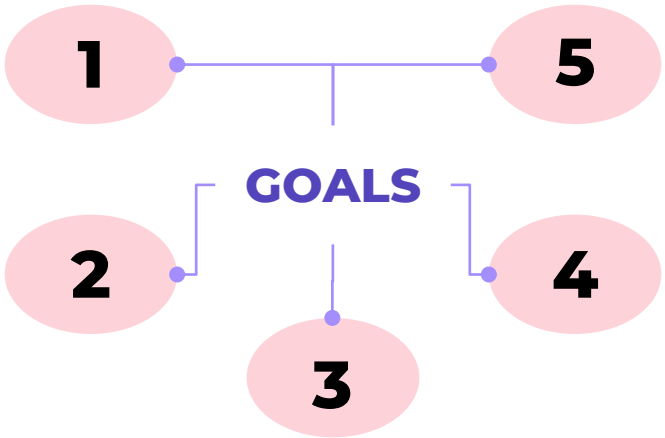
Contextualized



I watch TV
I wear watch

ELMo, BERT, etc

Research Objectives



STUDY

Study existing bias in static & contextualized word embeddings

EXPLORE

Look for strategies to identify bias

DETECT

Detecting bias in transformer models

MITIGATE

Continue pretraining to mitigate bias

COMPARE

Compare before and after debiasing

Related Work

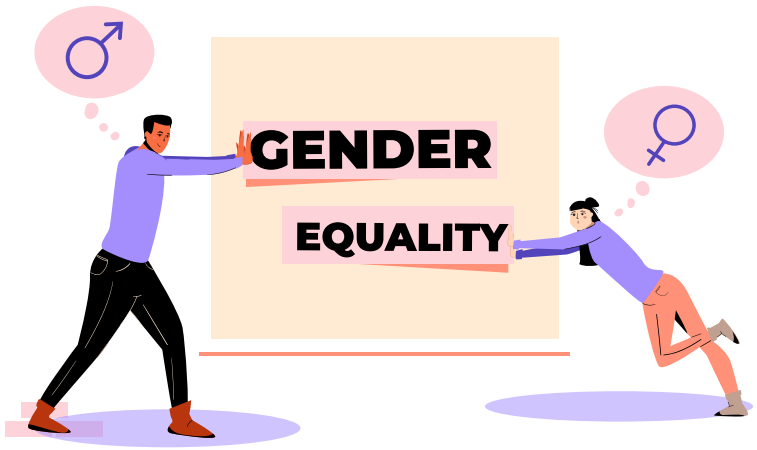
- Bolukbasi et al. (2016) performed the direct bias test by measuring the cosine similarity of the gender direction and occupations.

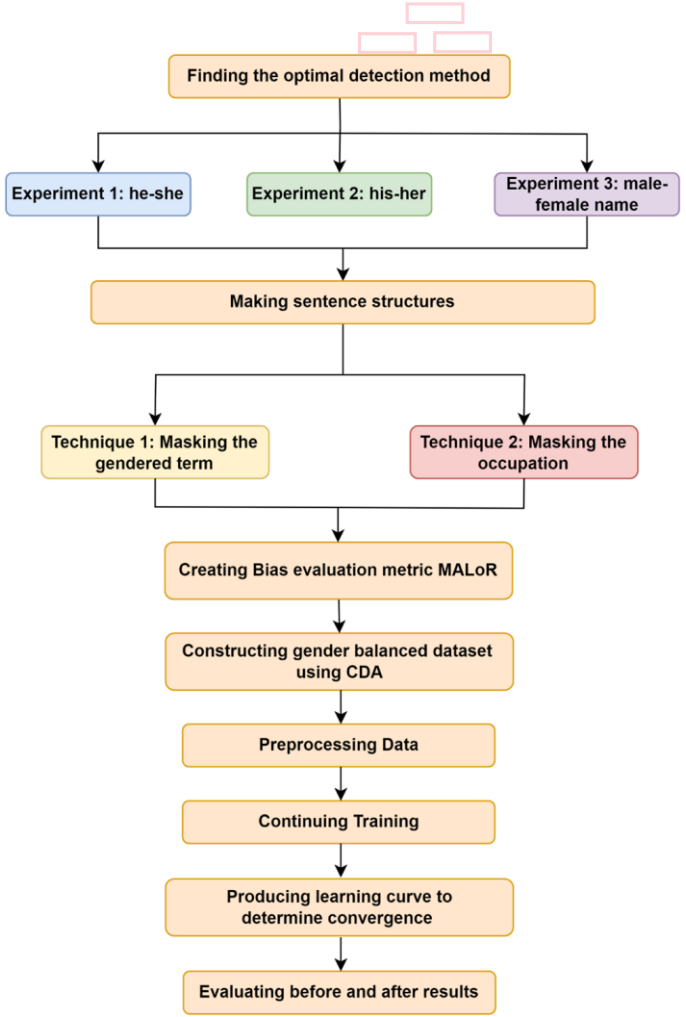
$$\overrightarrow{\text{man}} - \overrightarrow{\text{woman}} \approx \overrightarrow{\text{computer programmer}} - \overrightarrow{\text{homemaker}}$$

- Kurita et al. (2019) used masking technique to find the probabilities of the masked tokens being replaced by other tokens. They used log normalized probability to represent bias.

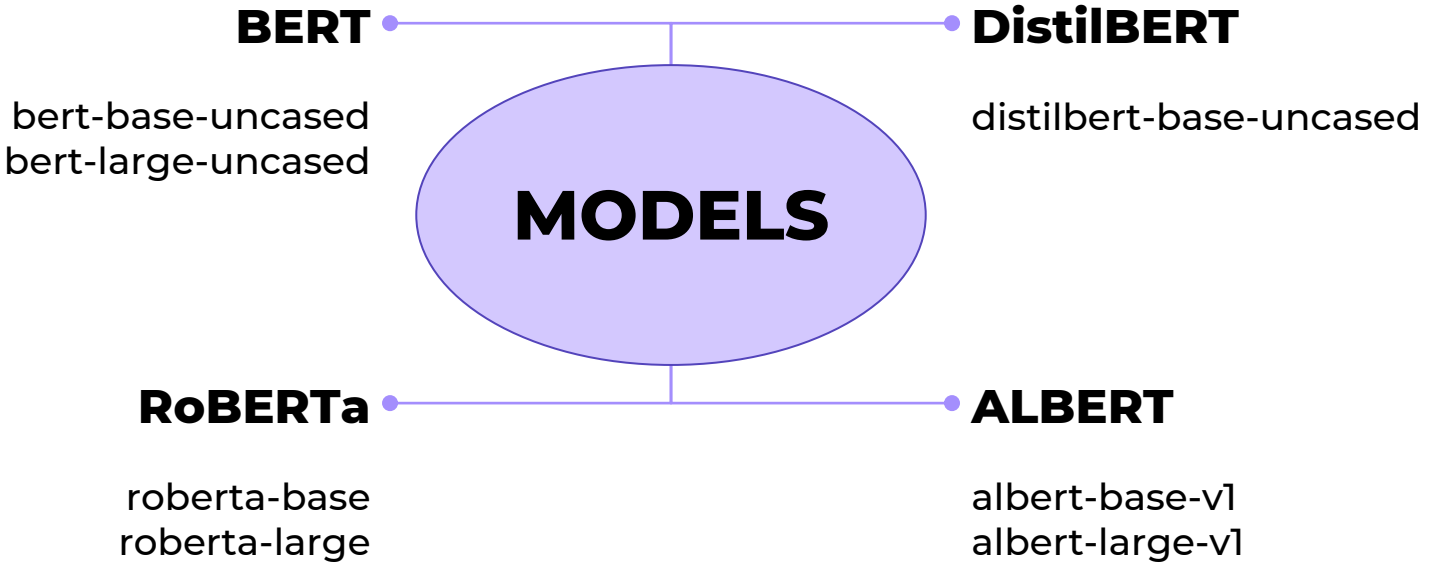
- Bartl et al. (2020) fine tuned BERT on the GAP corpus and used Kurita et al's (2019) log normalized probability to show the debiased result.

Work Flow


GENDER
EQUALITY

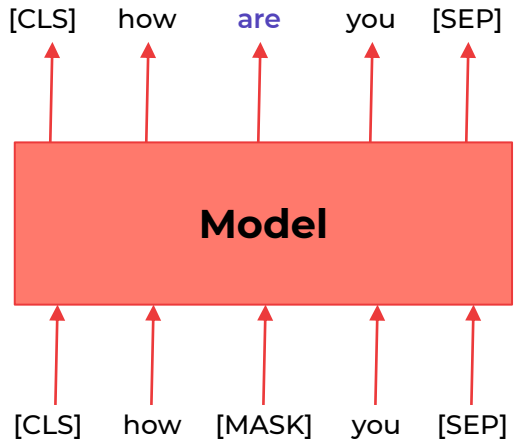

Finding the optimal detection method
Experiment 1: he-she
Experiment 2: his-her
Experiment 3: male-female name
Making sentence structures
Technique 1: Masking the gendered term
Technique 2: Masking the occupation
Creating Bias evaluation metric MALoR
Constructing gender balanced dataset using CDA
Preprocessing Data
Continuing Training
Producing learning curve to determine convergence
Evaluating before and after results

Explored Models



Masked Language Modeling (MLM)

These models are trained using Masked Language Modeling (MLM). We used this concept of MLM to detect gender bias.



- Select 15% of the tokens
- Replace 80% of the selected tokens with [MASK]
- Replace 10% of the selected tokens with a random word
- Keep the rest 10% of the selected tokens as it is

MASKING

He/She

[MASK] is a good engineer/nurse
he/she is a good [MASK]

His/Her

[MASK] dream is to be a good engineer/nurse
his/her dream is to be a good [MASK]

Male/Female names

My friend [MASK] is a good engineer/nurse
My friend Jack/Rose is a good [MASK]

String : [MASK] is a good engineer. | he : 0.599079012 ; she : 0.030344748

String : he is a good [MASK]. | engineer : 0.000271102 ; nurse : 0.000167707

He/She

- [MASK] dreams of being a full-time [OCC].
- [MASK] has always wanted to be a passionate [OCC].
- ..... (48)
- [MASK] is determined to be a successful [OCC].

His/her

- [MASK] dream is to become a full-time [OCC].
- [MASK] dreams of being a good [OCC].
- ..... (48)
- [MASK] is determined to be a successful [OCC].

Male/Female names

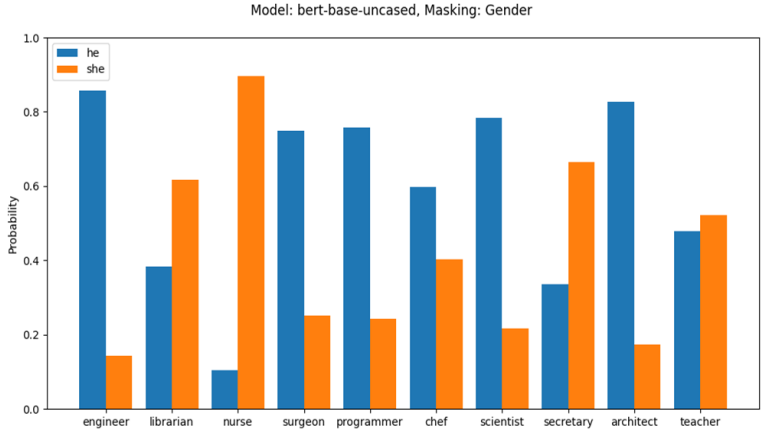
- My friend [MASK] dreams of being a full-time [OCC].
- My friend [MASK] has always wanted to be a passionate [OCC].
- ..... (48)
- My friend [MASK] is determined to be a successful [OCC].

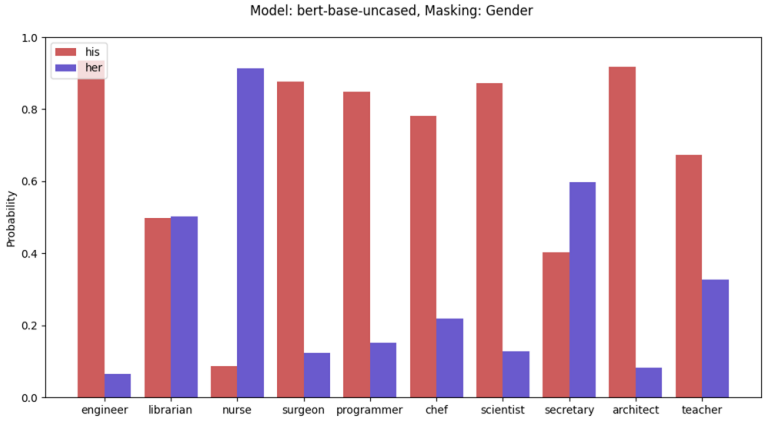
occupation = engineer, librarian, nurse, surgeon, programmer, chef, scientist, secretary, architect, teacher, ... (50)

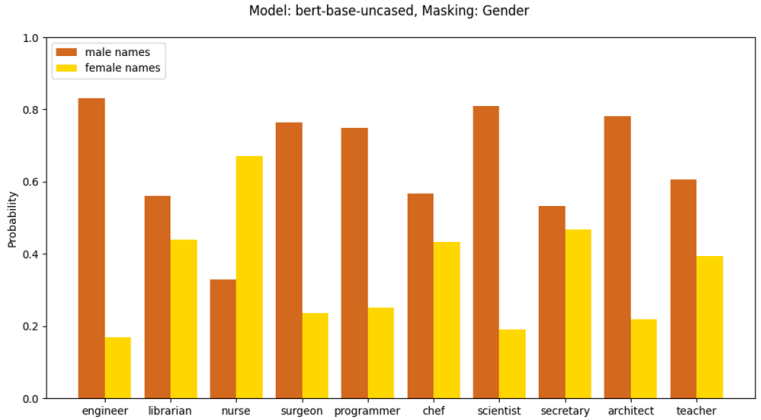
male names = Michael, David, James, John, Robert, William, Richard, Thomas, Christopher, Joseph ..... (19)

female names = Jennifer, Linda, Patricia, Susan, Mary, Sarah, Jessica, Elizabeth, Karen, Nancy, ... (19)

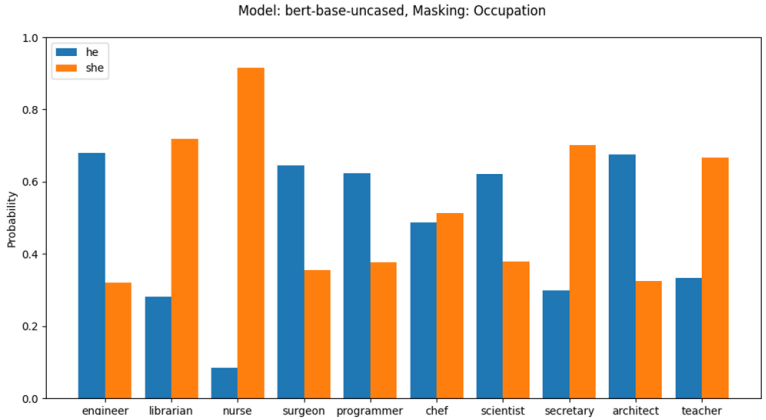
Masking Gender

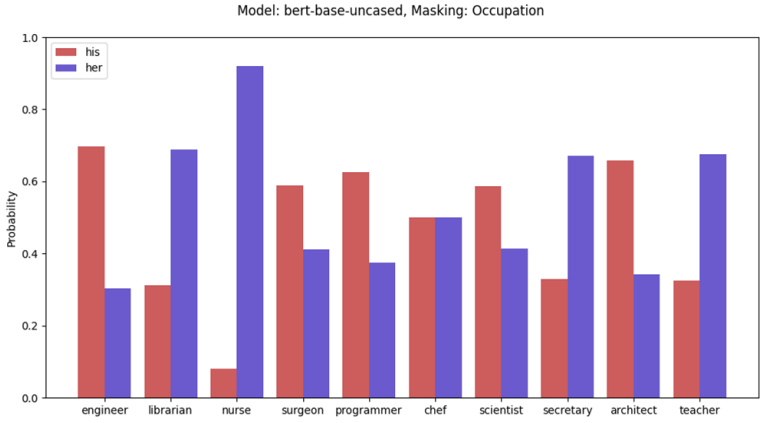


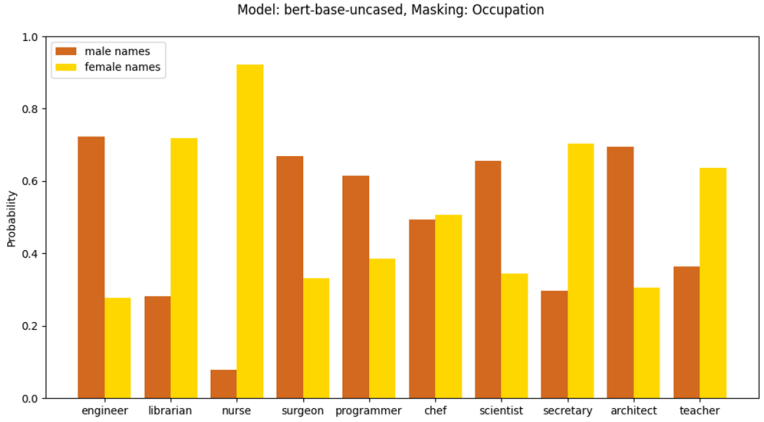




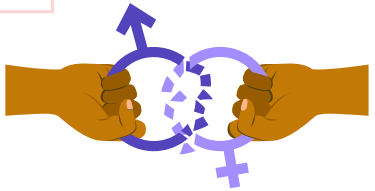
Masking Occupation







Evaluation Metric - MALoR



$$\frac{1}{M}\sum_{j\in \text{ occ}}\left|\left(\frac{1}{N}\sum_{i\in \text{ sent}}\log_2\frac{P(male_term)_{ij}}{P(female_term)_{ij}}\right)\right|$$

Here, sent=sentence structures used, occ=occupational words used. N=no. of sentence structures. M=no. of occupational words used.

He/She

male_term = he
female_term = she

His/Her

male_term = his
female_term = her

Male/Female names

male_term = avg of 29 male names
female_term = avg of 29 female names

MALoR Scores of Original Models

Model	he-she	his-her	male-female name
bert-base-uncased	1.27	2.51	1.37
bert-large-uncased	1.98	2.55	1.82
distilbert-base-uncased	0.633	2.09	0.603
roberta-base	1.64	1.581	N/A
roberta-large	0.790	1.81	N/A
albert-base-v1	0.619	2.58	N/A
albert-large-v1	0.250	2.26	N/A

Dataset For Mitigation

Source : WMT18 and WMT15 - News Corpus, News Commentary and other sources

Method : CDA applied to sentences containing gender terms and an occupational word from a set of 60 occupations.

01 He/She

Total Sentences : 6,848

Example :
1. "**She** is a capable teacher."
2. "**He** is a capable teacher."

02 His/Her

Total Sentences : 6,424

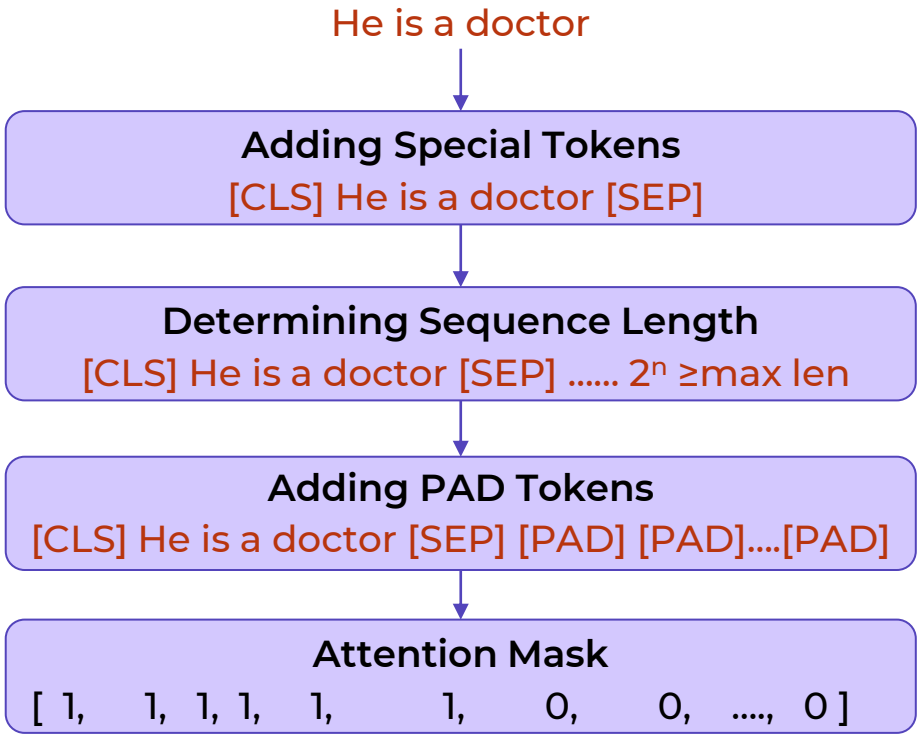
Example :
1."**Her** ambition is to be a lawyer."
2."**His** ambition is to be a lawyer."

03 Male/Female Names

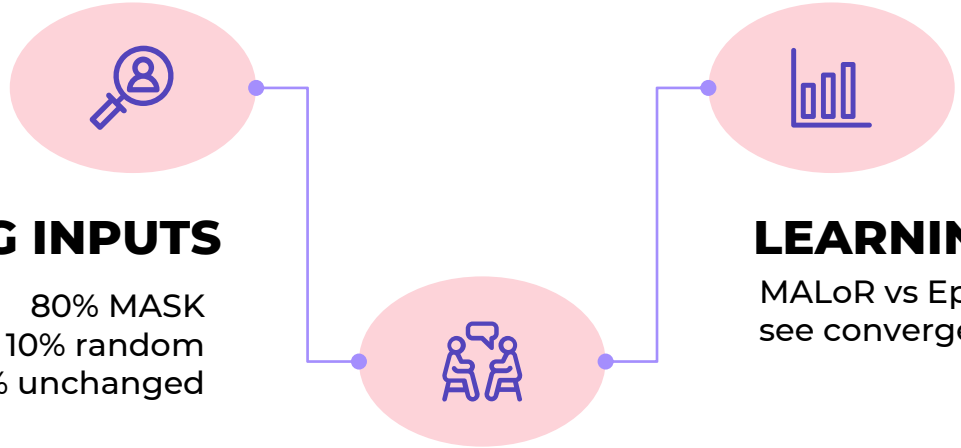
Total Sentences : 6,288

Example :
1. "**Emily** was a doctor."
2."**Stephen** was a doctor."

Preprocessing Data



Continue Pretraining



MASKING INPUTS

80% MASK
10% random
10% unchanged

LEARNING CURVE

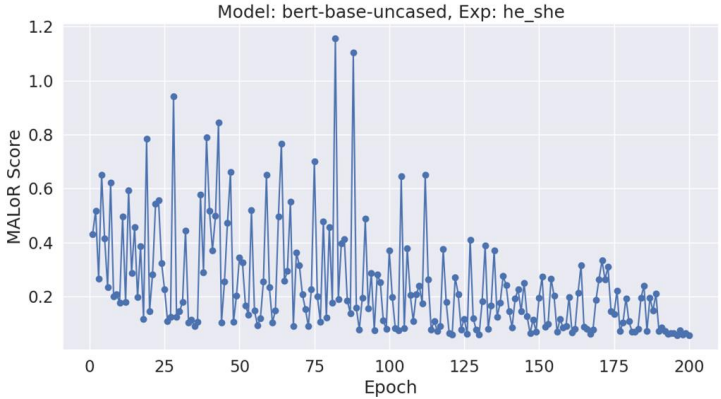
MALoR vs Epochs to
see convergence

HYPERPARAMETERS

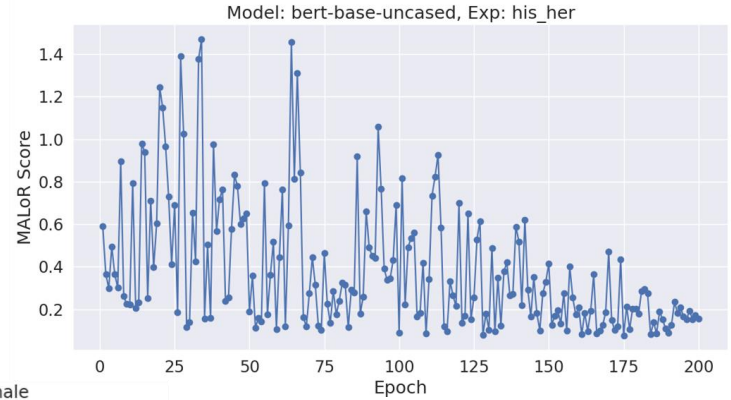
Learning rate, AdamW Optimizer,
Linear Scheduler, Batch Size

Learning Curves

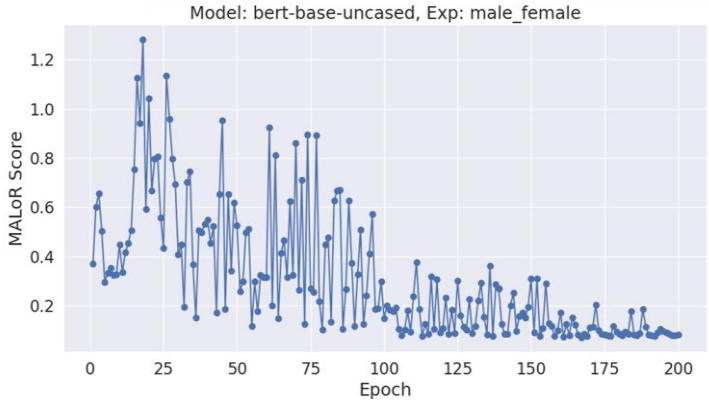
he-she



his-her



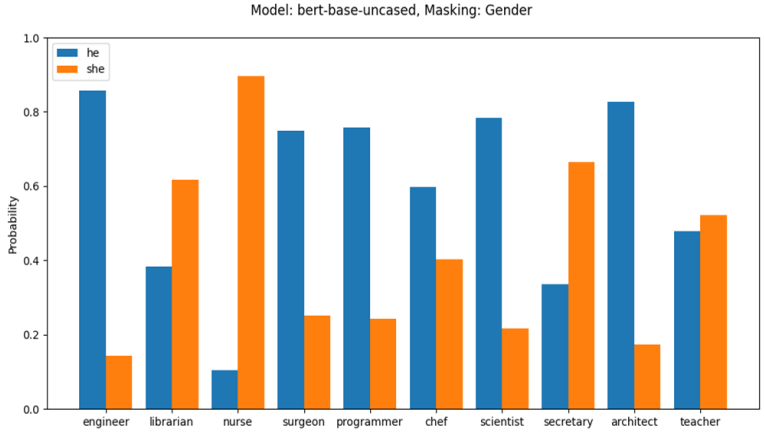
Male-female name

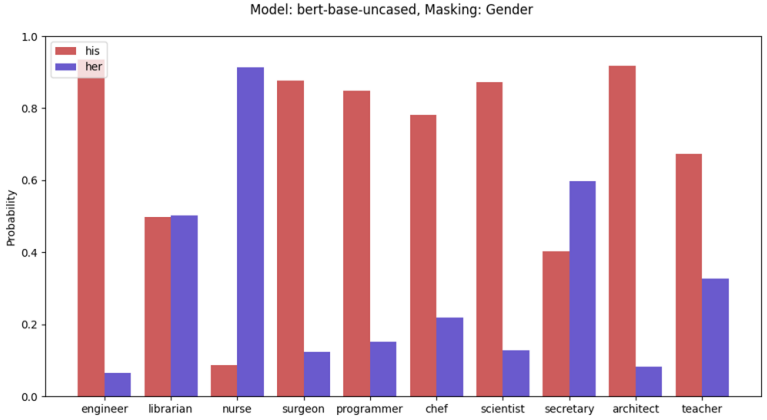


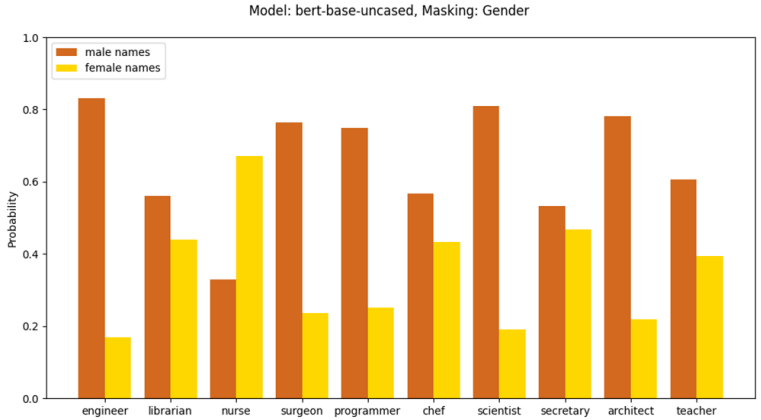
MALoR Scores of Debiased Models

Model	he-she (Before)	he-she (After)	his-her (Before)	his-her (After)	male-female name (Before)	male-female name (After)
bert-base-uncased	1.27	0.0803 ± 0.0147	2.51	0.488 ± 0.224	1.37	0.418 ± 0.242
bert-large-uncased	1.98	0.0594	2.55	0.610	1.82	0.338
distilbert-base-uncased	0.633	0.126 ± 0.0606	2.09	0.179 ± 0.0684	0.603	0.416 ± 0.157

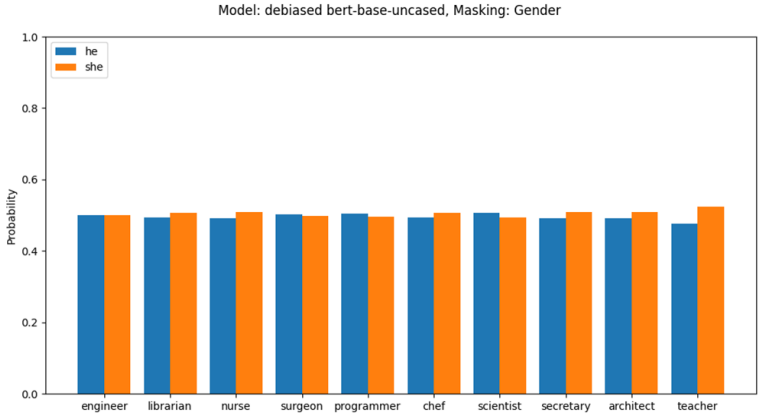
Masking Gender (Before)

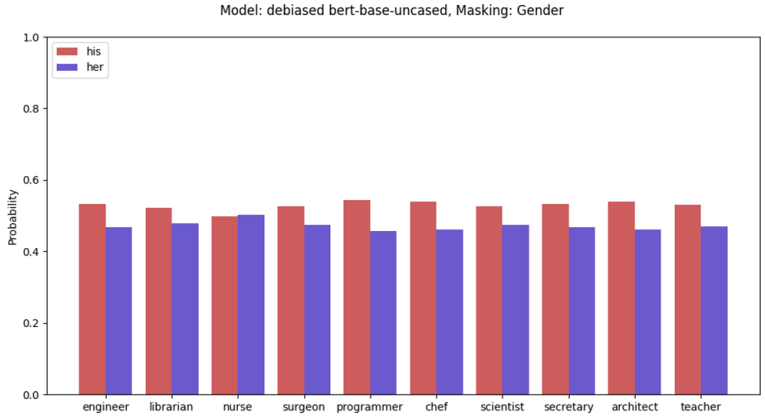


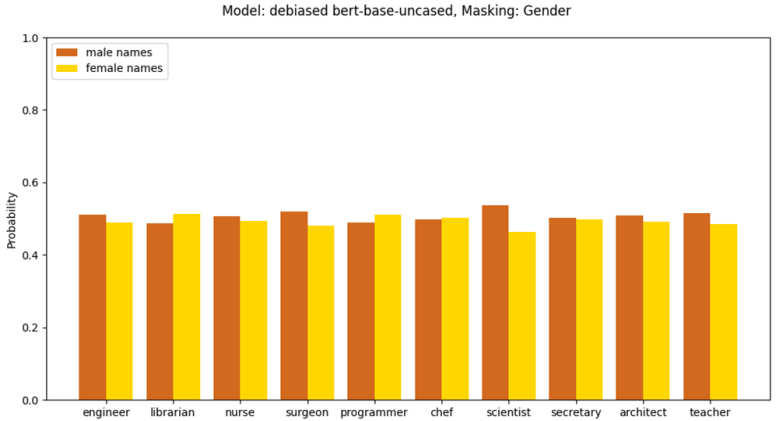




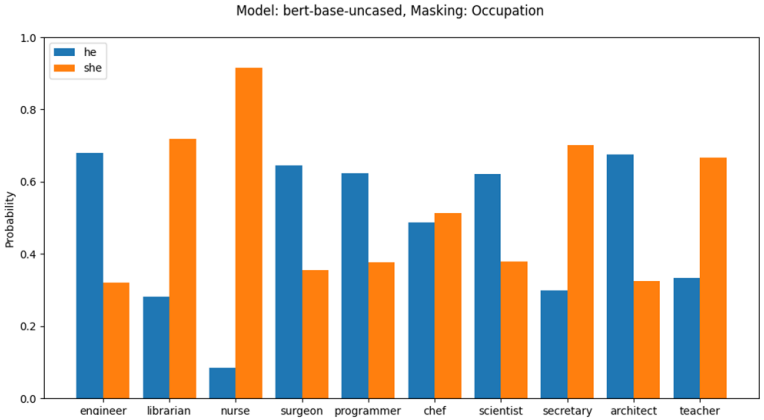
Masking Gender (After)

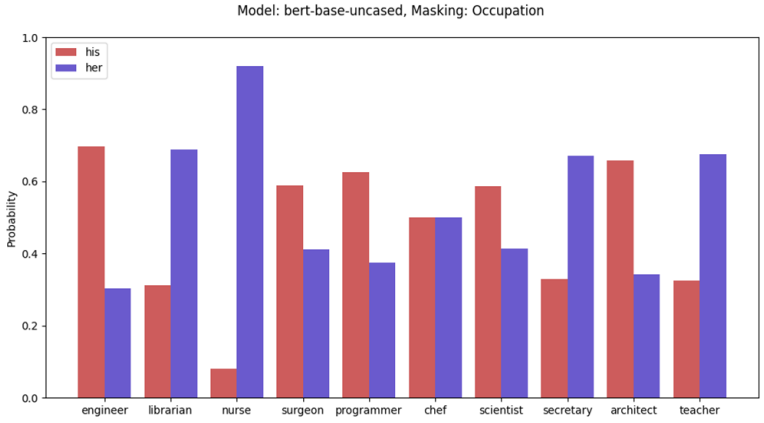


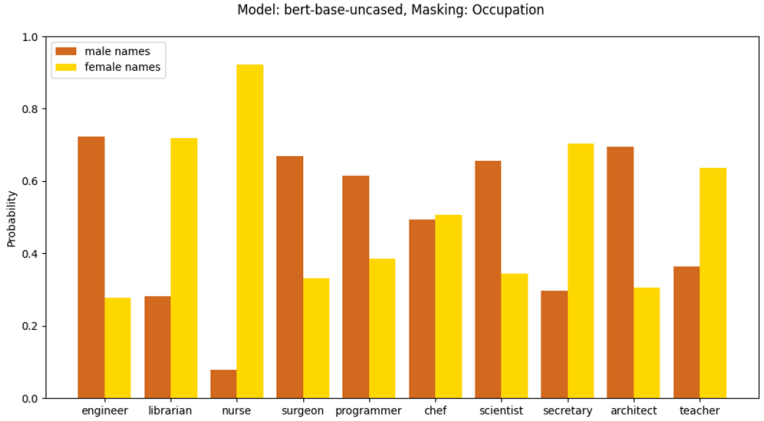




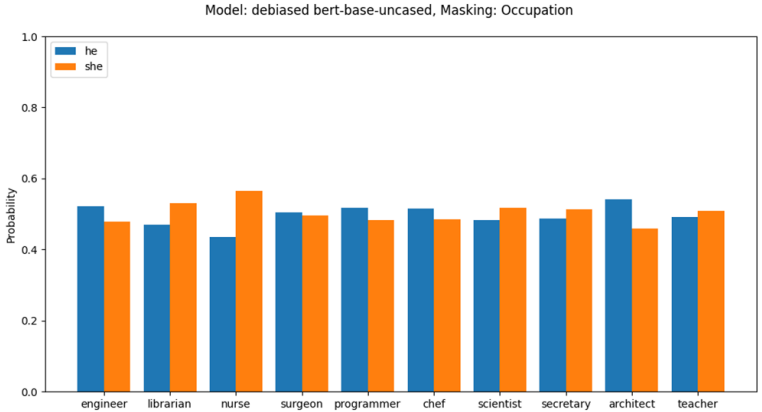
Masking Occupation (Before)

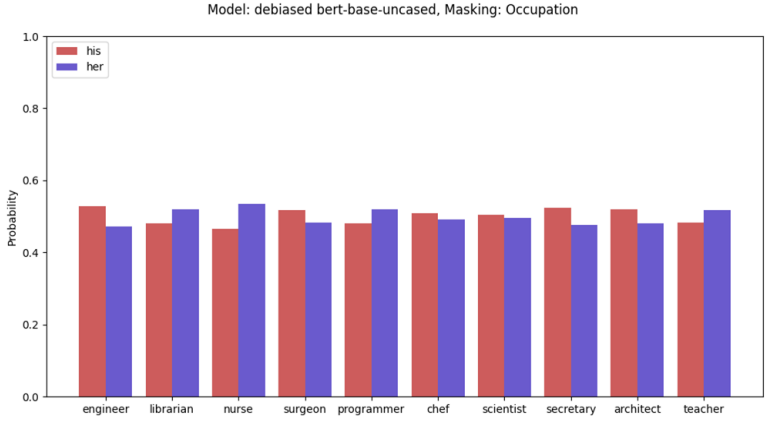


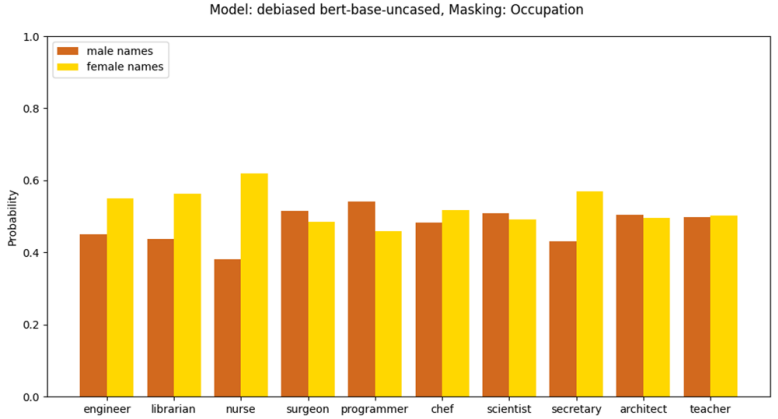




Masking Occupation (After)







Quality Assurance

We performed Sentiment Analysis from GLUE's SST-2 task with both original and debiased BERT. The objective is to categorize each sentence as "positive" or "negative".

- **Null Hypothesis (H_0):** The mean accuracy of debiased BERT is not significantly different from the mean accuracy of the original BERT on the SST-2 task.
- **Alternative Hypothesis (H_a):** The mean accuracy of debiased BERT is significantly different from the mean accuracy of the original BERT on the SST-2 task.

Significance level ($\alpha = 0.05$)
p-value = 0.5671627142

p-value > α , we fail to reject the null hypothesis. There is no statistically significant difference between the "before" and "after" values at the 0.05 significance level.

Conclusion

01

Findings

- Not discovering bias through traditional methods which are used in static embeddings.
- Thorough analysis of bias in different models using masking probability
- Found differences of vocabulary/tokenization process in these models
- DistilBERT having bad convergence

02

Contributions

- Creation of an evaluation metric MALoR, applicable to various models
- Successful debiasing method using CDA to create gender balanced datasets
- Demonstration of effective debiasing on 3 experiments
- Assurance of BERT's Quality using Sentiment Analysis

Conclusion

03

Limitations

- Limited focus on gendered pronouns and names
- Time-consuming debiasing process
- Incomplete comparison of all models due to different vocabularies/tokenization process
- Unable to calculate mean and std of MALoR Scores for BERT Large due to hardware and time constraint

04

Future Work

- Expand analysis to include other gender terms
- Mitigate biases in models with different vocabularies/tokenization process
- Calculate mean and std of MALoR Scores for BERT Large
- Determine better convergence for DistilBERT

References

T. Bolukbasi, K.-W. Chang, J. Zou, V. Saligrama, and A. Kalai, “Man is to computer programmer as woman is to homemaker? debiasing word embeddings,” NIPS, 2016. doi: 10.48550/arXiv.1607.06520. eprint: https://doi.org/ 10.48550/arXiv.1607.06520. [Online]. Available: https://doi.org/10.48550/ arXiv.1607.06520.

K. Kurita, N. Vyas, A. Pareek, A. W. Black, and Y. Tsvetkov, Measuring bias in contextualized word representations, 2019. arXiv: 1906.07337 [cs.CL].

M. Bartl, M. Nissim, and A. Gatt, “Unmasking contextual stereotypes: Measuring and mitigating bert’s gender bias,” CoRR, vol. abs/2010.14534, 2020. arXiv: 2010.14534. [Online]. Available: https://arxiv.org/abs/2010.14534.